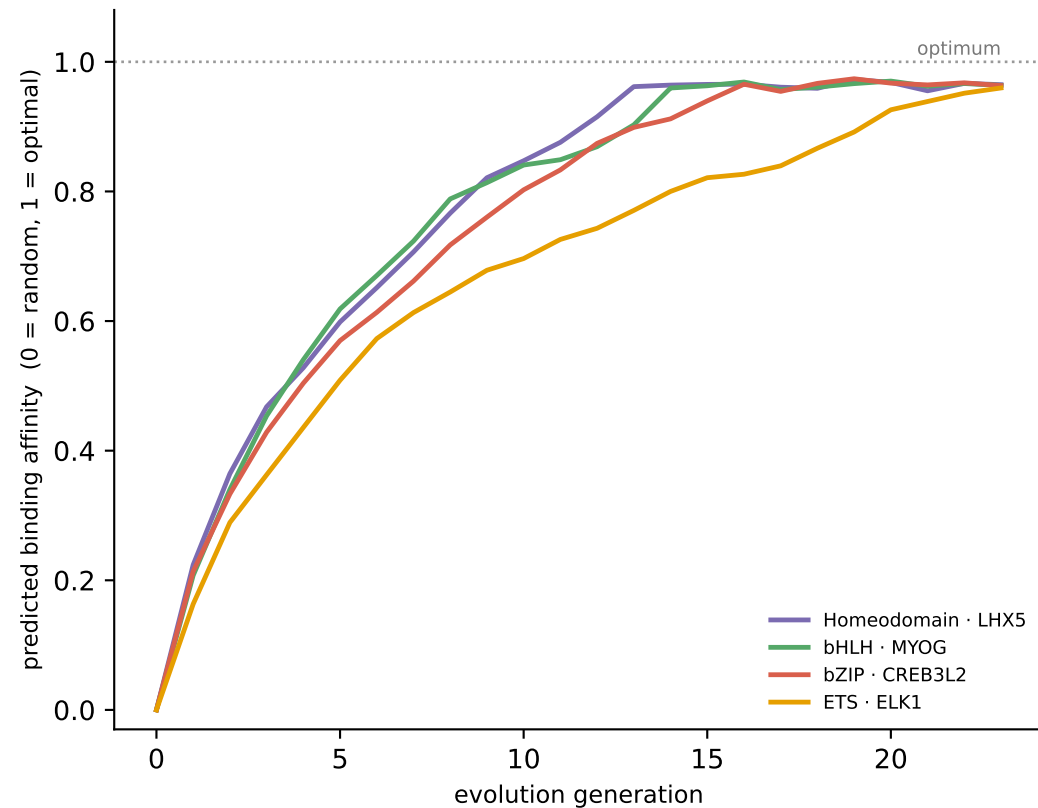


### a In-silico SELEX converges from random DNA



### b Random DNA → recovered consensus

start: random DNA (flat, ~0 bits)

evolved TAATTA · LHX5 (homeodomain,  $r=0.985$ )

**T A A T T A**

### c Each factor recovers its own motif

evolved CAGCTG · MYOG (bHLH,  $r=1.0$ )

**C A G C T G**

evolved ACGTGG · CREB3L2 (bZIP,  $r=0.985$ )

**A C G T G G**

evolved ACTTCGGT · ELK1 (ETS,  $r=0.938$ )

**A C T T C C G G T**